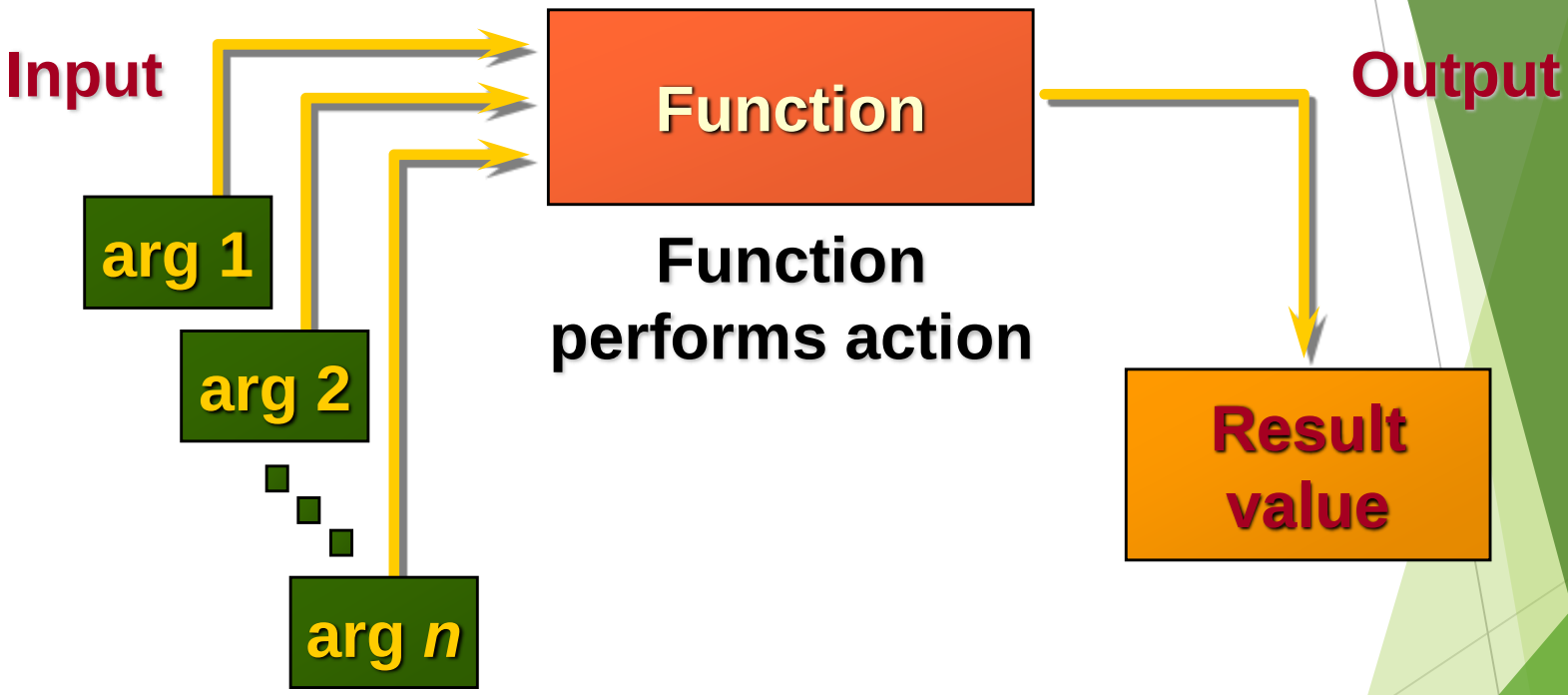
FUNCTIONS

Objectives

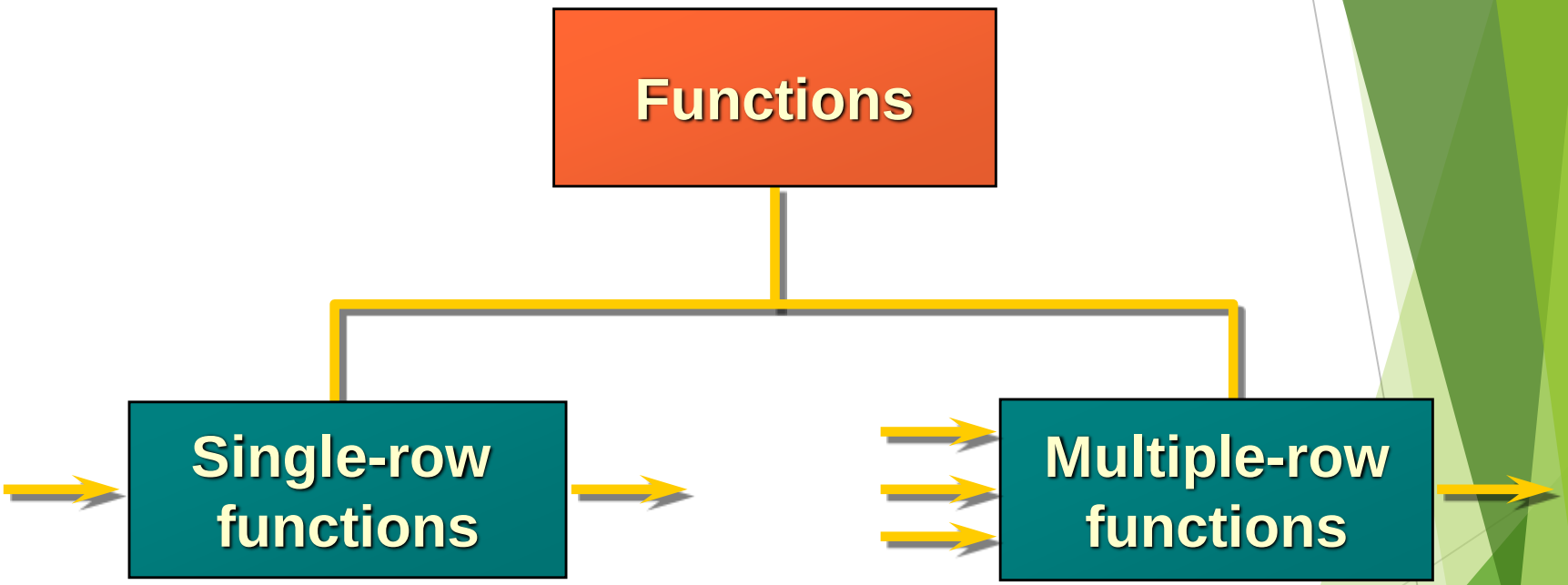
- After completing this lesson, you should be able to do the following:
  - Describe various types of functions available in SQL
  - Use character, number, and date functions in SELECT statements
  - Describe the use of conversion functions



SQL Functions



Types of SQL Functions

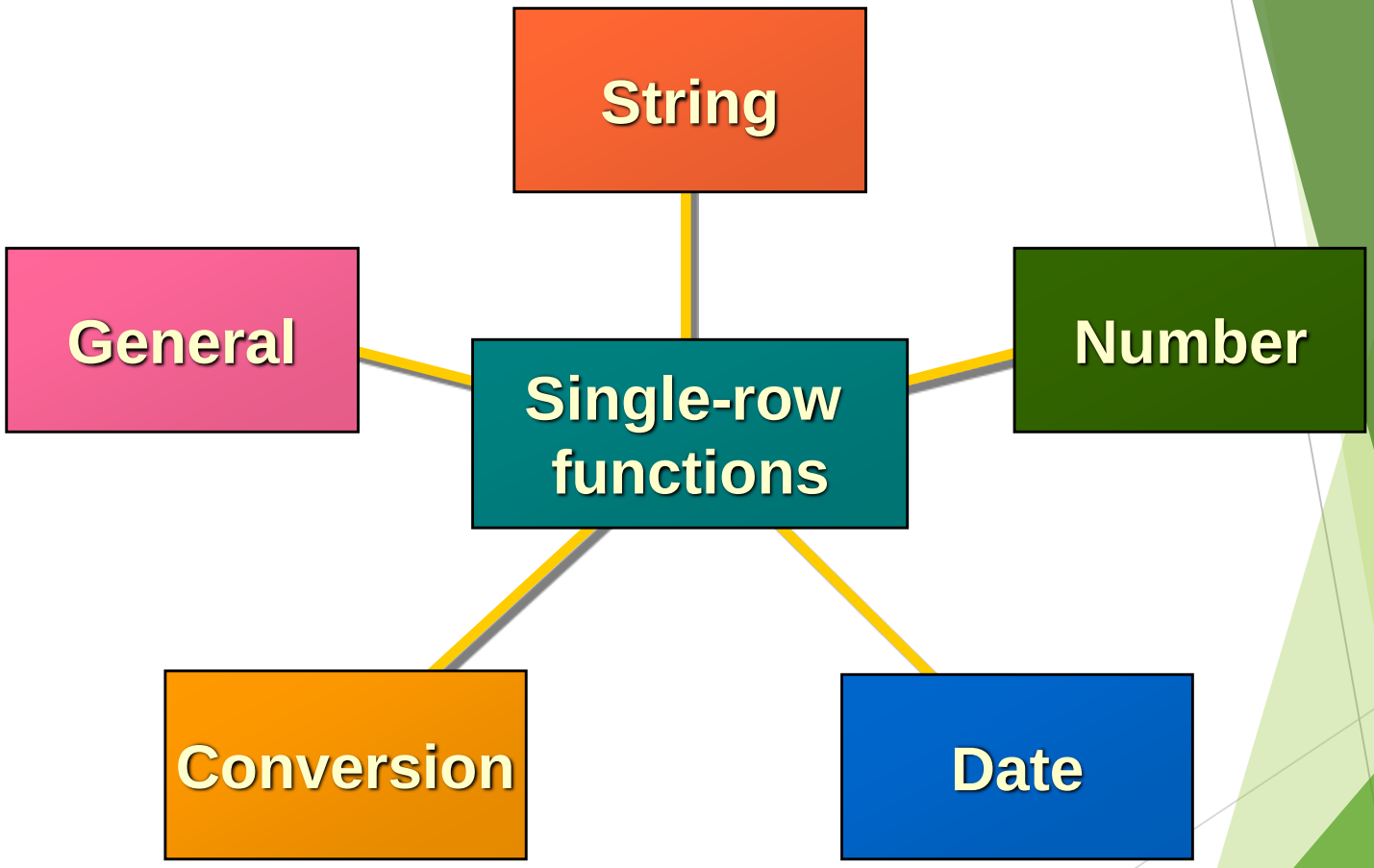


Single-Row Functions

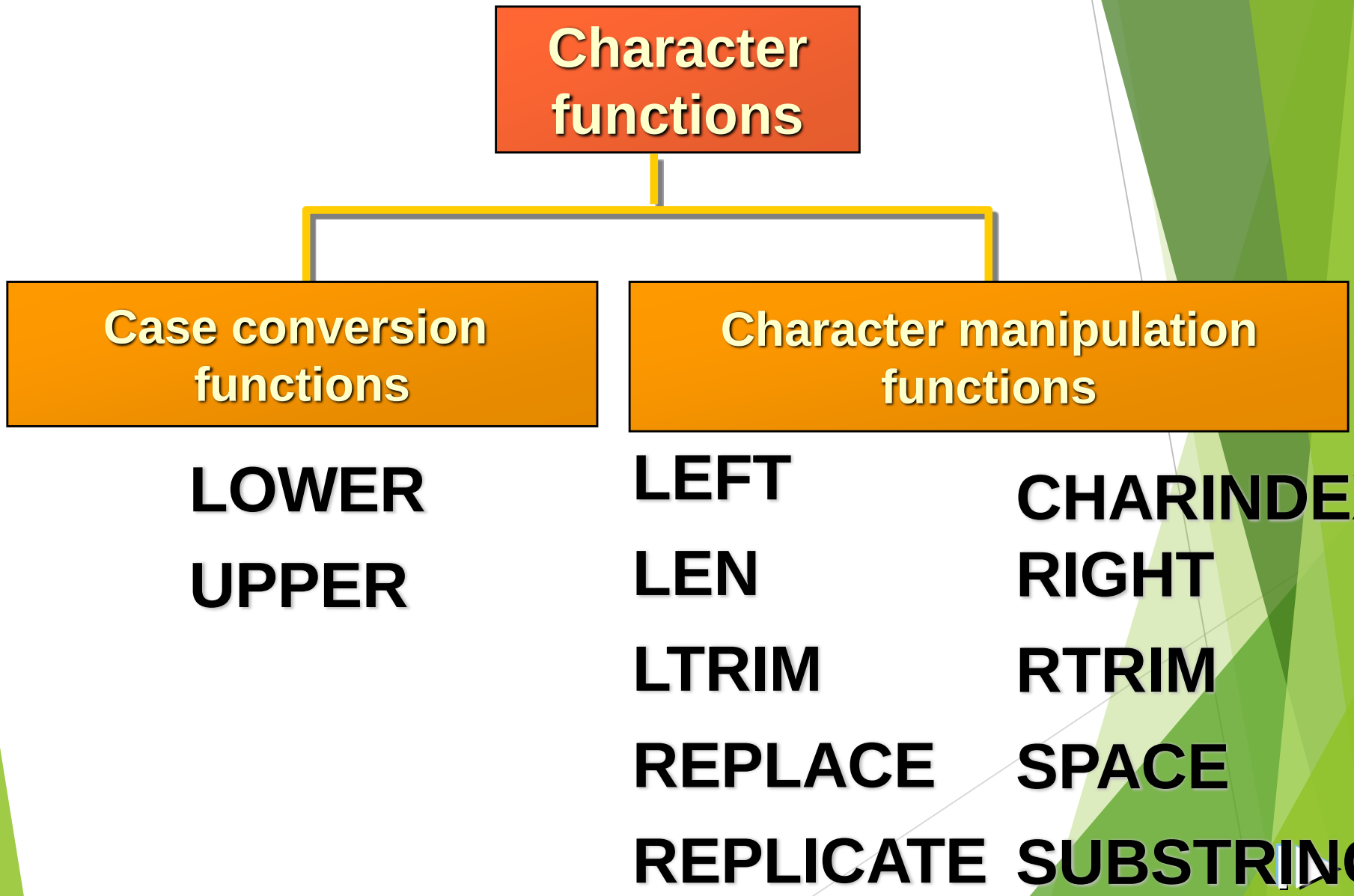
- Manipulate data items
- Accept arguments and return one value
- Act on each row returned
- Return one result per row
- May modify the datatype
- Can be nested

```
function_name (column|expression, [arg1, arg2,...])
```

Single-Row Functions



String Functions



Case Conversion Functions

- Convert case for character strings

Function	Result
LOWER('SQL Course')	sql course
UPPER ('SQL Course')	SQL COURSE

Case Conversion Functions

- Display the employee ID, name for employee King.

```
SELECT EmployeeID,
       UPPER(LastName)  Lname,
       LOWER(FirstName)  Fname
FROM HR.employees
WHERE LastName = 'Davis'
```

EmployeeID	Lname	Fname
7	DAVIS	sara

LEN

- Returns the number of characters of the specified string expression

LEN(*char_expr*)

LEN

```
SELECT DISTINCT title,
       LEN(title)length
FROM HR.employees
```

```
title                                    length
---------------------                    -------
CEO                                        3
Sales Manager                             13
Sales Representative                      20
Vice President, Sales                     21
```

LEFT

- Returns the part of a character string starting at a specified number of characters from the left

LEFT(*char_expr,integer_expr*)

LEFT

```
SELECT title,
       LEFT(title,4)
FROM HR.employees
```

title	(no column name)
CEO	CEO
Sales Manager	Sale
Sales Representative	Sale
Vice President, Sales	Vice

RIGHT

- Returns the part of a character string starting at a specified number of characters from the right.

RIGHT(*char_expr,integer_expr*)

RIGHT

```
SELECT title,
       RIGHT(title,4)
FROM HR.employees
```

title	(no column name)
CEO	CEO
Sales Manager	ager
Sales Representative	tive
Vice President, Sales	ales

LTRIM

- Returns a character expression after removing leading blanks.

```
LTRIM (char_expr)
```

LTRIM

```
SELECT LTRIM('     abcde')
```

```
-------------
abcde
```

RTRIM

- Returns a character expression after removing trailing spaces.

```
RTRIM (char_expr)
```

RTRIM

```
SELECT RTRIM('abcde    ')
```

```
-------------
abcde
```

REPLACE

- Replaces all occurrences of the second given string expression in the first string expression with a third expression.

REPLACE ('*str_expr1*', '*str_expr2*', '*str_expr3*')

str_expr1 **- string to be searched**
str_expr2 **– string try to find**
str_expr3 **– replacement string**

REPLACE

```
SELECT REPLACE('abcdefghicde',
'cde','xxx')
```

```
----------------
abxxxfghixxx
```

REPLICATE

- Repeats a character expression for a specified number of times.

REPLICATE (char_*expr* , *integer_expr*)

REPLICATE

```
SELECT REPLICATE(FirstName,2)
FROM hr.employees
```

```
(no column  name)
----------------
SaraSara
DonDon
JudyJudy
YaelYael
...
```

SUBSTRING

- Returns part of a character, binary, text, or image expression

SUBSTRING(*expression*, *start*, *length*)

expression **- string**
start **– beginning position of the substring to be extracted**
length **– length of the string extracted**

SUBSTRING

```
SELECT FirstName,
     SUBSTRING(FirstName,2,5)
FROM HR.employees
```

FirstName	(no column name)
Sara	ara
Don	on
Judy	udy
Yael	ael
...	

CHARINDEX

- Returns the starting position of a string specify

CHARINDEX(*expr1,expr2 [,start]*)

expr1 **- string to be found**
expr2 **– string to be searched**
start **– expression at which the search starts**

CHARINDEX

```
SELECT DISTINCT title,
       CHARINDEX('ale',title)
FROM HR.employees
```

title	(no column name)
CEO	0
Sales Manager	2
Sales Representative	2
Vice President, Sales	18



String Manipulation Functions

- Manipulate character strings

Function	Result
LEFT('Good', 2)	Go
RIGHT ('Good', 2)	od
LEN('String')	6
LTRIM (' Good')	Good
RTRIM ('String ')	String

String Manipulation Functions

- Manipulate character strings

Function	Result
REPLACE('Good', 'o', 'a')	Gaad
REPLICATE ('Good', 2)	GoodGood
CHARINDEX('o','Good')	2
SUBSTRING('String',3,4)	ring

Mathematical Functions

Numeric Functions

- **ABS**
- **POWER**
- **CEILING**
- **FLOOR**
- **ROUND**

Mathematical Functions

- **ABS**: Return the absolute value of the number

  ABS(-1.0) 1.0

- **POWER**: **Returns the value of the given expression to the specified power**

  POWER(2,2) 4
  POWER(3,4) 81

Mathematical Functions

- **CEILING: Returns the smallest integer greater than, or equal to, the given numeric expression.**

CEILING(123.45)	124
CEILING(-123.45)	-123
CEILING(0.0)	0

Mathematical Functions

- **FLOOR**: Returns the largest integer less than, or equal to, the given numeric expression

FLOOR(123.45)	123
FLOOR(-123.45)	-124
FLOOR(0.0)	0

Mathematical Functions

- **ROUND**: **Returns a numeric expression, rounded to the specified length or precision**

ROUND(748.58, -1)	750.00
ROUND(748.58, -2)	700.00
ROUND(748.58, -3)	1000.00
ROUND(123.4545, 2)	123.4500
ROUND(123.45, -2)	100.00

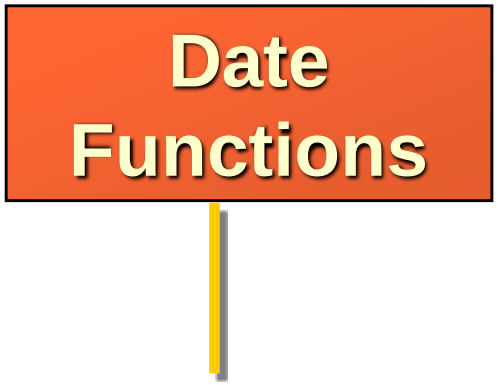


Using the ROUND Function

```
SELECT ROUND(45.923,2) 'ROUND(45.923,2)' ,
       ROUND(45.923,0) 'ROUND(45.923,0)',
       ROUND(45.923,-1) 'ROUND(45.923,-1)'
```

ROUND(45.923,2)	ROUND(45.923,0)	ROUND(45.923,-1)
45.92	46	50

Date Functions



DATENAME

DATEPART

DATEADD

DATEDIFF

Date Function

- Sometimes we only want a portion of a date or we want to do date arithmetic

- To do this we will use the following functions:
  - datename
  - datepart
  - dateadd
  - datediff

Arithmetic with Dates

- Add or subtract a number to or from a date for a resultant *date* value.
- Subtract two dates to find the *number* of days between those dates.
- Add *hours* to a date by dividing the number of hours by 24.

GETDATE

- Returns the current system date and time in the Microsoft® SQL Server™ standard internal format for datetime values.

```
GETDATE ()
```

GETDATE

```
SELECT GETDATE()
```

```
-----------------------------
2009-07-30 16:09:26.170
```

DATEADD

- Returns a new datetime value based on adding an interval to the specified date.

```
DATEADD (datepart,number,date)
```

Datepart	Abbreviations	Datepart	Abbreviations
Year	yy, yyyy	Day	dd, d
quarter	qq, q	Week	wk, ww
Month	mm, m	Hour	hh
		minute	mi, n

DATEADD

```
SELECT Hiredate, DATEADD(dd,3,Hiredate) AddDays,
       DATEADD(mm,2,Hiredate)AddMonth,
       DATEADD(yy,1,Hiredate) AddYear
FROM HR.employees
```

Hiredate	AddDays	AddMonth	AddYear
2002-05-01	2002-05-04	2002-07-01	2003-05-01
2002-08-14	2002-08-17	2002-10-14	2003-08-14
2002-04-01	2002-04-04	2002-06-01	2003-04-01
...			

DATEDIFF

- Returns the number of date and time boundaries crossed between two specified dates

```
DATEDIFF(datepart,startdate,enddate)
```

Datepart	Abbreviations	Datepart	Abbreviations
Year	yy, yyyy	Day	dd, d
quarter	qq, q	Week	wk, ww
Month	mm, m	Hour	hh
		minute	mi, n

DATEDIFF

```
SELECT Hiredate,
  DATEDIFF(dd,Hiredate,GETDATE()) DiffDays
  DATEDIFF(mm,Hiredate,GETDATE())DiffMon
  DATEDIFF(yy,HIREDATE,GETDATE()) DiffYea
FROM HR.employees
```

Hiredate	DiffDays	DiffMonth	DiffYear
2002-05-01	4115	135	11
2002-08-14	4010	132	11
2002-04-01	4145	136	11
…			

DATENAME

- Returns a character string representing the specified datepart of the specified date.

```
DATENAME(datepart,date)
```

DATENAME

```
SELECT DATENAME(mm, getdate())
    AS 'Month Name'
```

```
Month Name
----------
August
```

DATEPART

- Returns an integer representing the specified datepart of the specified date

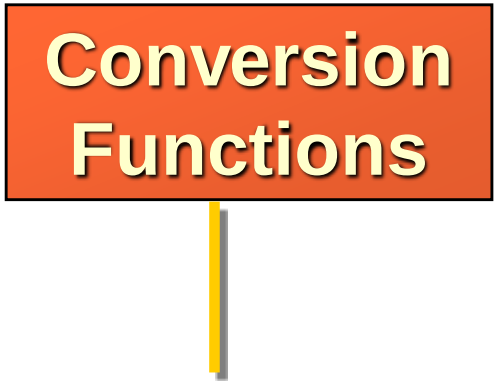
```
DATEPART(datepart,date)
```

DATEPART

```
SELECT DATEPART(mm, GETDATE()) AS 'Month',
       DATEPART(day, GETDATE()) AS 'Day',
       DATEPART(year, GETDATE()) AS 'Year'
```

Month	Day	Year
8	6	2013

Conversion Functions



CONVERT

CAST

CAST and CONVERT

- CAST and CONVERT provide similar functionality
- Explicitly converts an expression of one data type to another

CAST

CAST(*expr AS datatype*)

- *expression*

  Is any valid Microsoft® SQL Server™ expression

- ***data_type***

  Is the target system-supplied data type



CAST

```
SELECT birthdate,
       CAST(birthdate AS varchar(11))
FROM HR.employees
```

birthdate	(No column name)
1958-12-08	Dec 8 1958
1962-02-19	Feb 19 1962
1973-08-30	Aug 30 1973
1947-09-19	Sep 19 1947

…

What is Wrong with this Statement?

```
SELECT ProductName+' unit price is '
    + UnitPrice 'Products Price'
FROM Production.products
```

Error Message

Cannot convert a char value to money

CAST

```
SELECT ProductName+' unit price is ’
       + CAST(UnitPrice AS varchar(20))
        ‘Product”s Price’
FROM Production.products
```

```
Product's Price
--------------------------------------------

Product HHYDP unit price is 18.00
Product RECZE unit price is 19.00
Product IMEHJ unit price is 10.00
Product KSBRM unit price is 22.00
```

Convert

CONVERT(*datatype[(length)], expr[,style]*)

- ***data_type***

  Is the target system-supplied data type

- *length*

  Is an optional parameter of nchar, nvarchar, char, varchar, binary, or varbinary data types



Convert

CONVERT(*datatype[(length)], expr[,style]*)

- ***expression***

  Is any valid Microsoft® SQL Server™ expression

- *style*

  Is the style of date format used to convert datetime or smalldatetime data to character data



Convert

```
SELECT CONVERT(varchar(20), birthdate,1)'mm/dd/yy',
       CONVERT(varchar(20), birthdate,3) 'dd/mm/yy',
       CONVERT(varchar(20), birthdate,6) 'dd mmm yy',
       CONVERT(varchar(11), birthdate,9) 'mmm dd yyyy'
FROM HR.employees
```

mm/dd/yy	dd/mm/yy	dd mmm yy	mmm dd yyyy
12/08/58	08/12/58	08 Dec 58	Dec 8 1958
02/19/62	19/02/62	19 Feb 62	Feb 19 1962
08/30/73	30/08/73	30 Aug 73	Aug 30 1973
09/19/47	19/09/47	19 Sep 47	Sep 19 1947
...			

```
SELECT CONVERT( VARCHAR(19), GETDATE() )
```

Aug 6 2013 7:06PM

```
SELECT CAST(GETDATE()  AS VARCHAR(19))
```

Aug 6 2013 7:06PM

```
SELECT CONVERT( VARCHAR, GETDATE() )
```

Aug 6 2013 7:06PM

```
SELECT CAST(GETDATE()  AS VARCHAR)
```

Aug 6 2013 7:06PM

```
SELECT CONVERT(VARCHAR(10), GETDATE(),101 )
```

```
08/06/2013
```

```
SELECT MONTH(GETDATE() )
```

```
8
```

```
SELECT DAY( GETDATE() )
```

```
6
```

```
SELECT YEAR(GETDATE())
```

```
2013
```

General Functions

